

Unit 5, Lesson 1: Scale Drawings

Benchmark

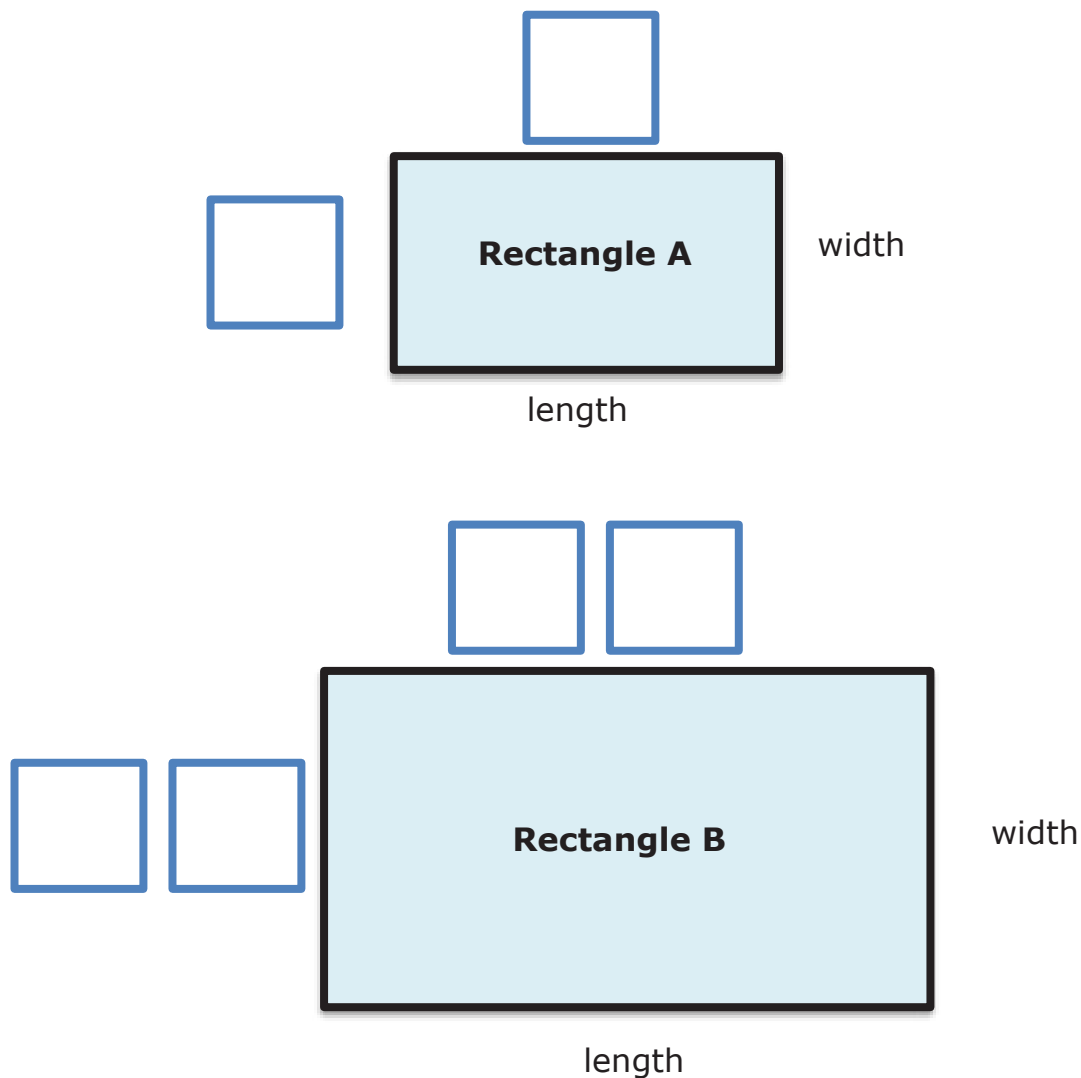
MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Learning Target

- ☐ I can draw a scaled figure on grid paper given a scale factor.

5.1.1 Warm-Up: Scaled Copies

Two rectangles are shown with their dimensions missing.



1. What are the lengths and widths of Rectangle A and Rectangle B that make the ratios $\frac{\text{width of Rectangle A}}{\text{width of Rectangle B}}$ and $\frac{\text{length of Rectangle A}}{\text{length of Rectangle B}}$ equal? Fill in the blue boxes using only the digits 0 – 9 so that each box has a different number inside.

5.1.2 Guided Instruction: Scale Drawings and Scale Factors

In the warm-up, the ratio of the widths of rectangles A and B is equivalent to the ratio of the lengths of rectangles A and B . Therefore, rectangle A is a scale model of rectangle B .

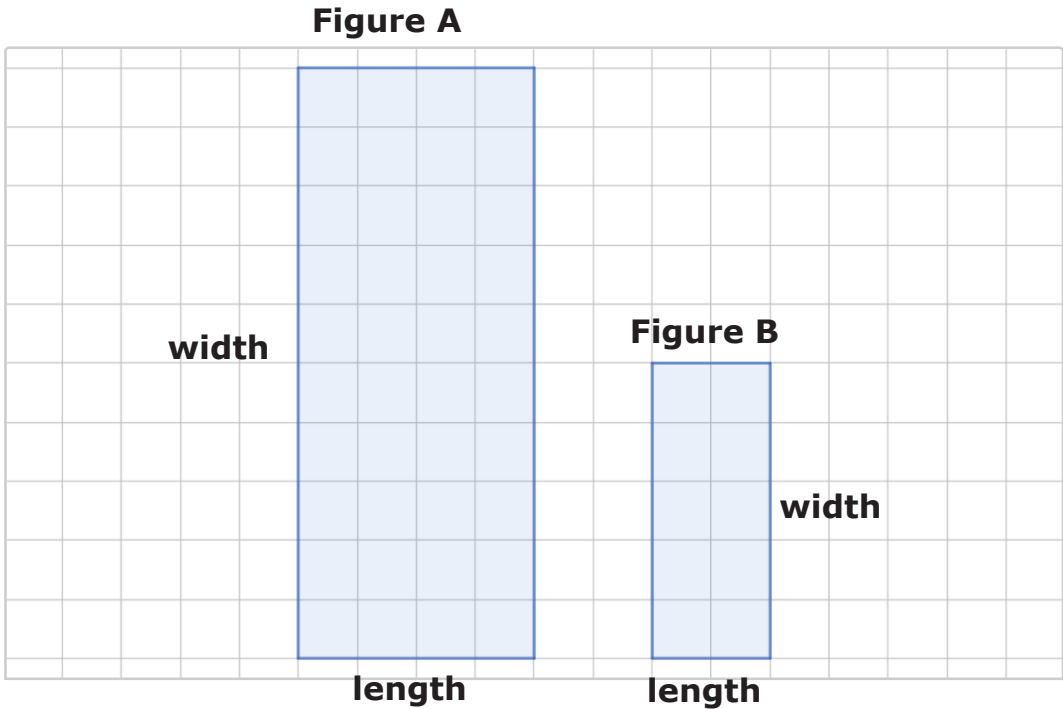


A **scale model** is a model or drawing based on a ratio of the dimensions for the model and the actual object it represents.

1. The keychain shown of the Eiffel Tower is a scale model of the real Eiffel Tower in Paris.
 - a. Sketch a drawing of the Eiffel Tower that is not a scale model.
 - b. Discuss with your partner why your drawing is not a scale model. Write a summary of your discussion.



2. Celeste drew an outline of a building for a diagram, and then drew a second model slightly smaller than her original drawing. Each gridline represents 1 centimeter (cm).



The length of Figure A and the length of Figure B can be written as a ratio because they are corresponding parts. This is also true for the widths of the figures.

Corresponding parts are parts of each figure that are in the same position in relation to the rest of each figure.

3. Complete the table for Figures A and B. Each gridline represents 1 cm.

	Length (cm)	Width (cm)
Figure A		
Figure B		
Ratio of Corresponding Parts $\frac{\text{Figure A}}{\text{Figure B}}$		

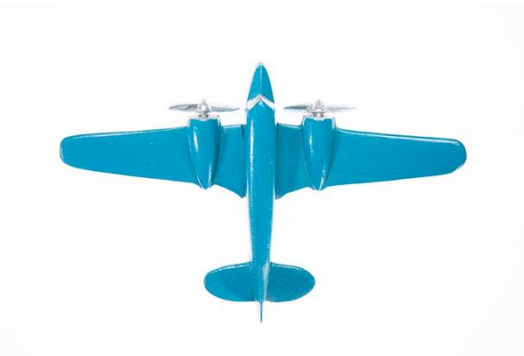
A model is a scale model if the ratios of the corresponding parts of the figures are equivalent. These ratios can be used to determine the scale factor that relates the figures.



A **scale factor** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.

4. Discuss with your partner what you notice about the scale factor that uses $\frac{\text{Figure A}}{\text{Figure B}}$ and the scale factor that uses $\frac{\text{Figure B}}{\text{Figure A}}$. Write a summary of your discussion.

Scale models can be scaled up or scaled down.

Scaled Down From the Original	Original	Scaled Up From the Original
		

5. Complete the statements about Figure A and Figure B.

The scale factor that relates Figure A to Figure B is _____ because each side length of Figure A is multiplied by this scale factor to get the side lengths of Figure B.

This is an example of scaling a figure

- ☐ up
- ☐ down

because the figure gets

- ☐ larger.
- ☐ smaller.

The scale factor that relates Figure B to Figure A is _____, which is an example of scaling the figure

- ☐ up
- ☐ down

because the figure gets

- ☐ larger.
- ☐ smaller.

5.1.3 Activity: Creating a Scale Drawing

Architectural drafters work in many industries to create drafts, or scale drawings, for different projects. They use specialized computer programs to create model drawings of buildings, wiring systems, topographical maps, bridges, and the electrical systems in many products.

Working with your small group, create a scale drawing of a specific room, area, or building. Each group will be assigned a different area as well as a different scale factor. If the rooms are not available, your group may choose from the landmarks below.

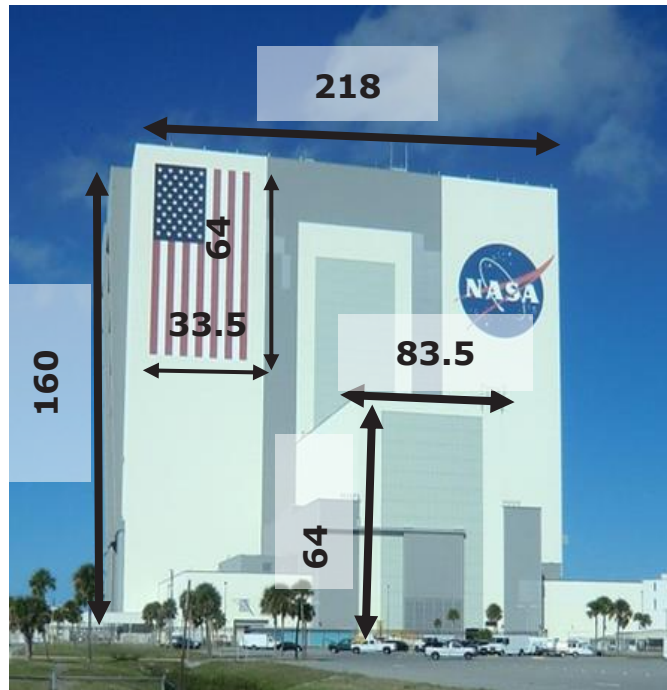
**The Singing Tower,
Bok Gardens
Lake Wales, FL**

**Measurements are in
meters (m)**



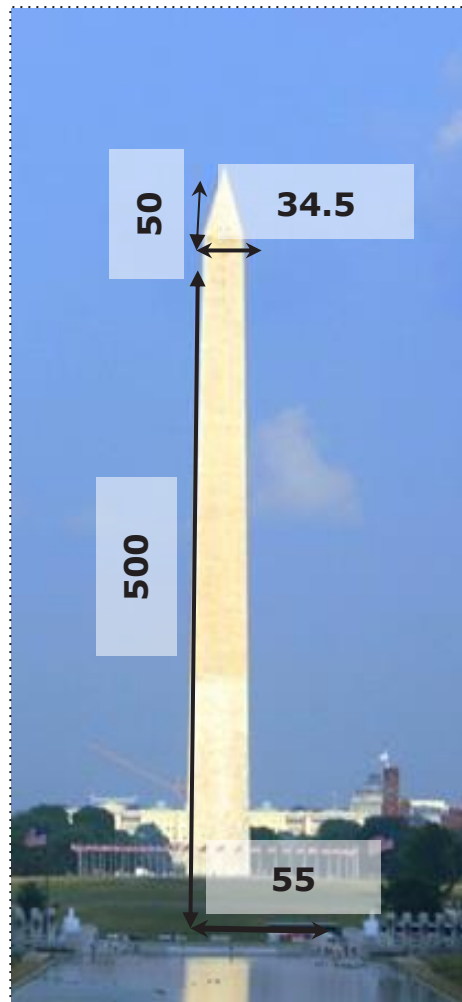
**NASA's Vehicle
Assembly Building
Kennedy Space
Center in FL**

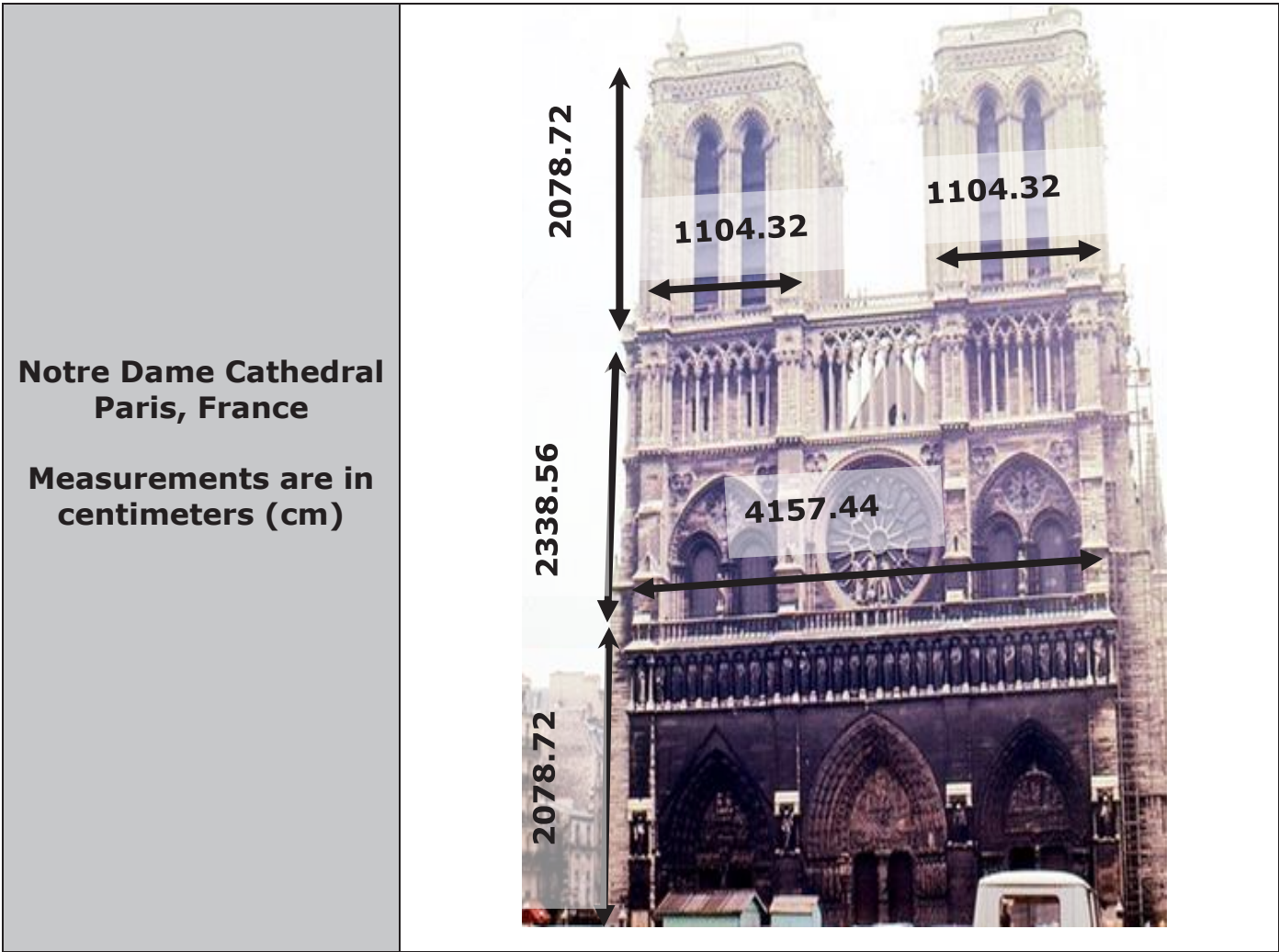
**Measurements are in
meters (m)**



**The Washington
Monument
Washington, D.C.**

**Measurements are in
feet (ft)**





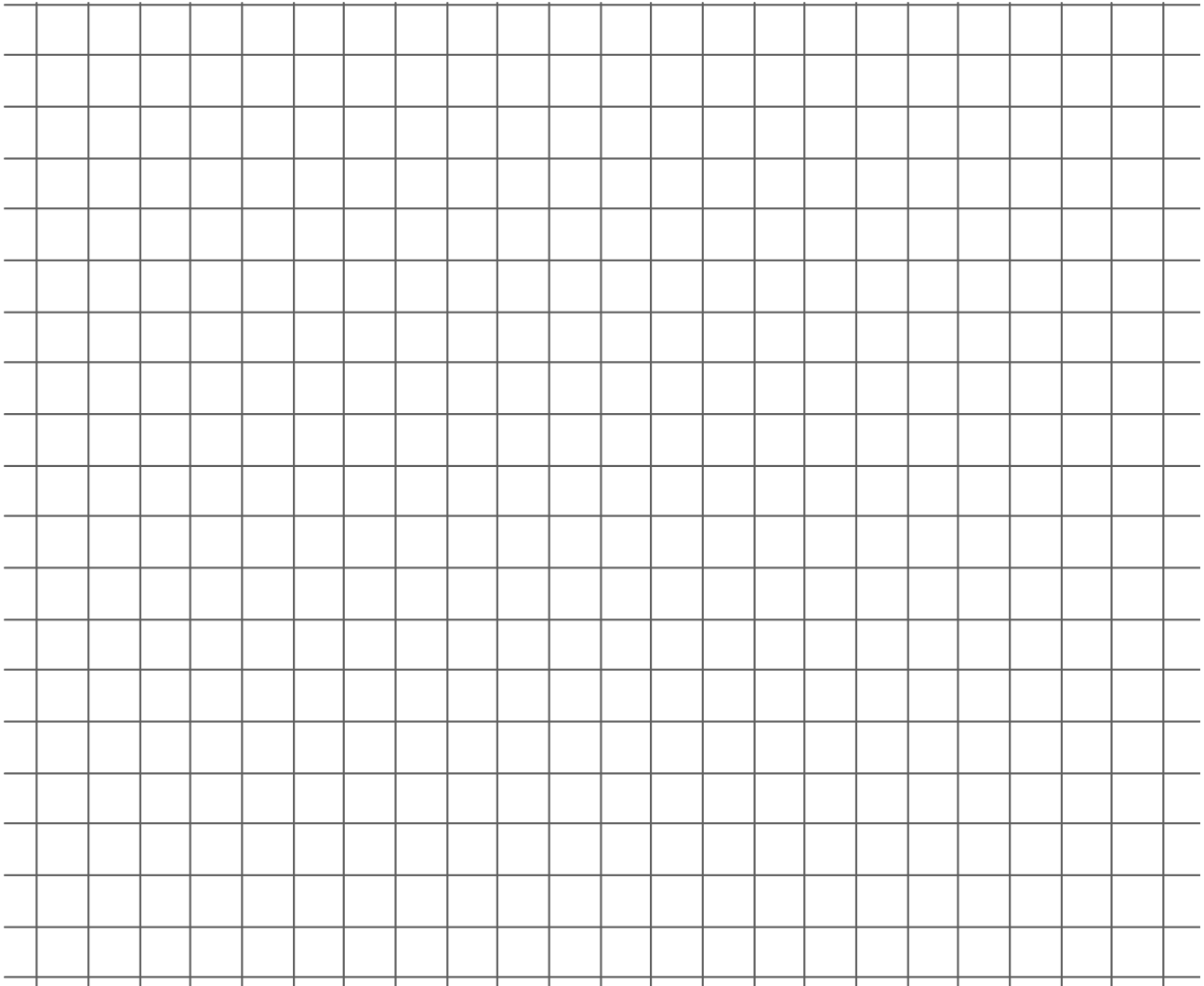
1. Complete the table based on your assignment. Sides five and six are provided if a room has alcoves or closets. If columns are not needed, leave them blank.

Room/area/landmark being measured: _____						
Scale factor: _____						
	Side 1	Side 2	Side 3	Side 4	Side 5	Side 6
Actual Measurement						
Scale Drawing Measurement						

2. Create your scale drawing. Complete the title and scale for your drawing. For example, one gridline might equal two meters.

Scale Drawing of _____

Scale: one gridline = _____



5.1.4 Collaboration: Numbered for Sharing

1. Number each member in your group. Partner with someone from a different group who has your same number. Discuss and record your answers to the following questions.

Questions	Reflections
Each of the scale factors assigned were fractions. Why do you think this makes sense for this assignment?	
How did your scale differ from your partner’s scale?	
How did the process of creating the scale model differ between you and your partner?	
How did the different scale factors affect the final drawing?	
Some groups may have turned their book sideways to create their scale drawing. Why do you think someone may have done that?	

5.1.5 Your Turn!

1. A polygon is shown. Each gridline represents one centimeter (cm).



- Draw a scaled copy of this polygon that has a perimeter of 27 cm.
- Explain how to determine the scale factor.

5.1.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate your understanding of today's learning target: "I can draw a scaled figure on grid paper given a scale factor."

